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Roll No

EC-5001-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Electromagnetic Field Theory

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1.
 - a) Explain Coulomb's Law.
 - b) Find the Equation for Energy Density in the Electrostatic Field.
2.
 - a) Find H due to a sheet of current.
 - b) Show that the relation between E and H in a uniform plane wave is $E/H = 120\Omega$.
3.
 - a) Explain Biot Savart's Law.
 - b) Write Maxwell's equation for time varying fields in both integral and differential form with statement.
4.
 - a) State and prove Gauss's Law.
 - b) Define gradient, curl and divergence with suitable examples. Write down their physical significance.
5. Explain:
 - i) Loss tangent
 - ii) Skin depth
 - iii) Surface current density

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6. a) Determine the magnetic field intensity $\vec{H}$ at the centre of a square current element. The length of each side is 2m and the current $I = 1.0\text{Amp}$.
b) Explain Faraday's law. Write the differential or point form of Faraday's law.
7. State and prove Biot-Savart's law in vector form magnetic field intensity.

OR

State and prove Ampere's circuital law in integral and differential vector form.

8. Write short notes on (any three)
 - a) Magnetic vector potential for sources in free space
 - b) Transmission line analogy
 - c) Uniqueness theorem
 - d) Frequency dispersive propagation
